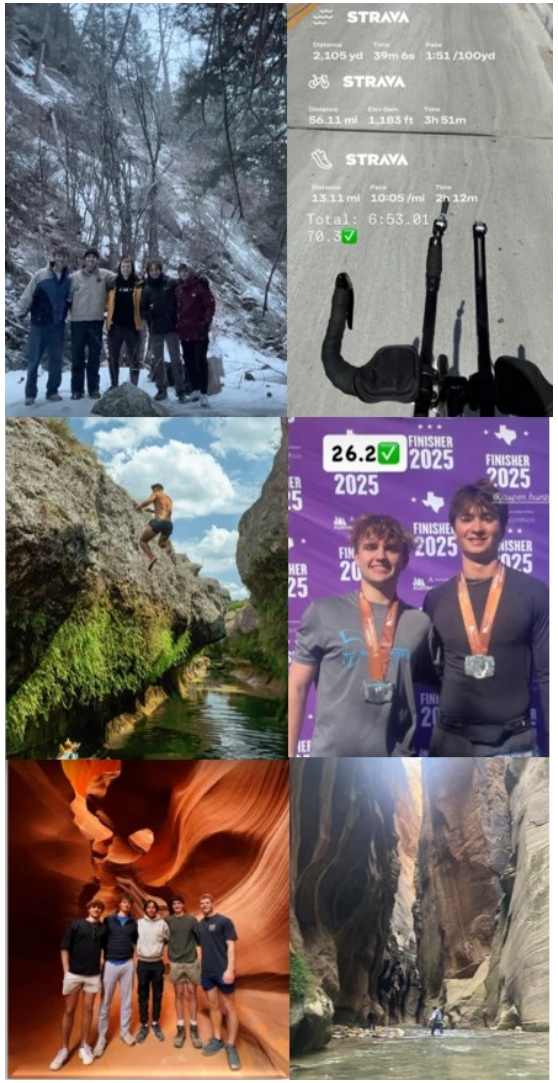
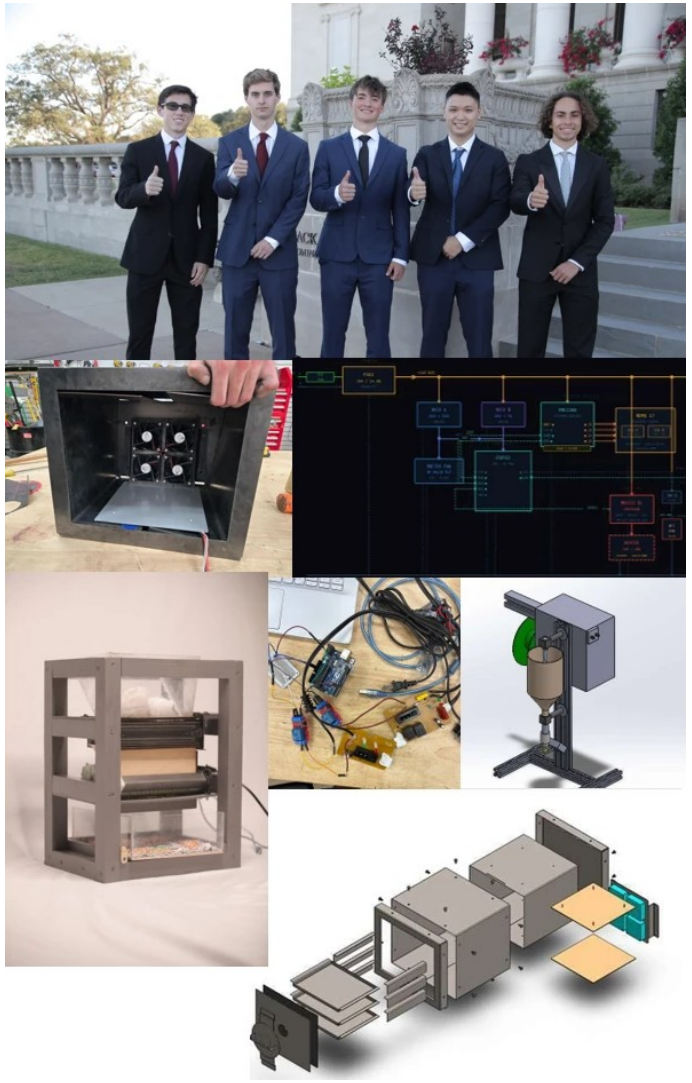
SpaceX Onsite

Jasper Buntinx

Mechanical Engineering, Texas A&M University

About Me


STRAVA
2,105 yd 39m 6s 1:51 /100yd
STRAVA
56.11 mi 1,183 ft 3h 51m
STRAVA
13.11 mi 10:05 /mi 2h 12m
Total: 6:53.01
26.2
FINISHER 2025



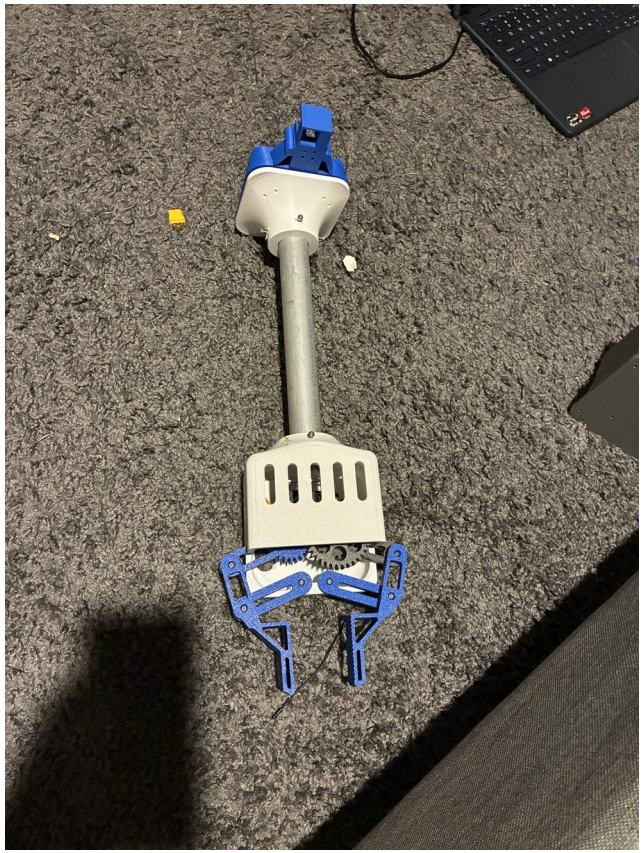

ONE ARMY
GLADIATOR DASH
$185K
RAISED FOR
STILL CREEK RANCH
FINISH

RoomCleaner

A four-cable ceiling robot designed to pick laundry off the floor

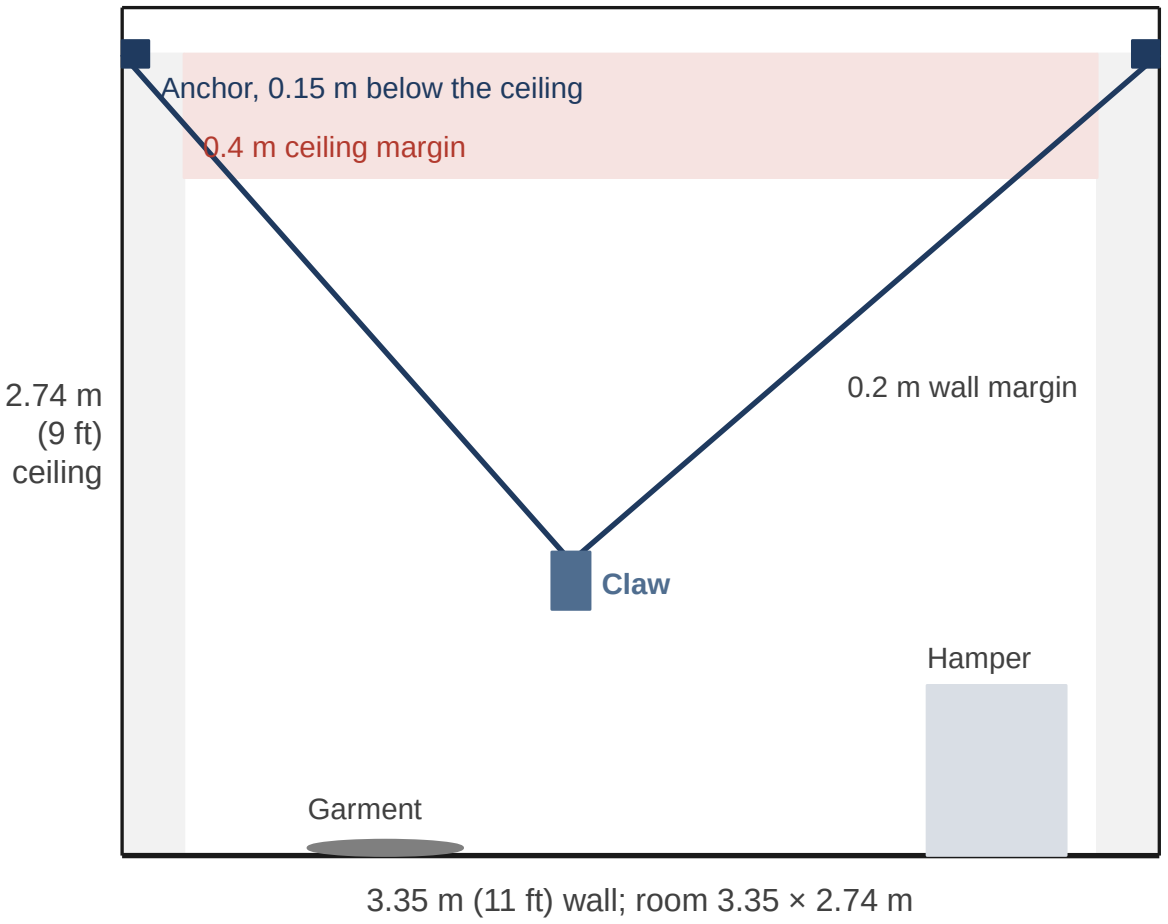


Wall station on the wall (built)



Claw (built)

Project Goal

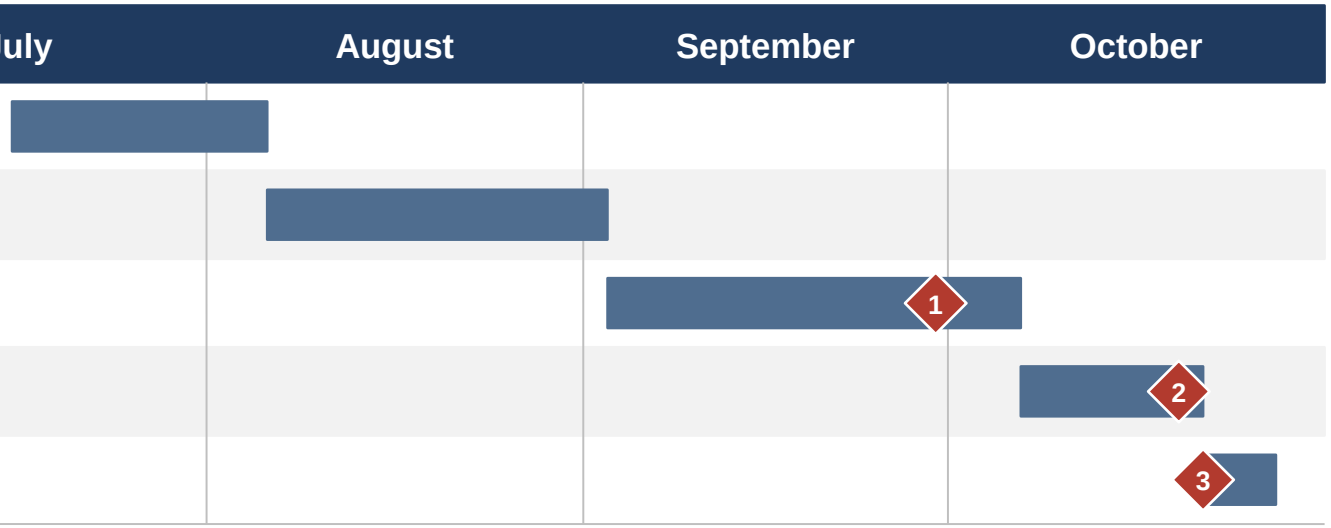


Room elevation to scale; seen from the side, the four cables appear as two

- Pick up laundry in a 3.35 × 2.74 m (11 × 9 ft) room under a 2.74 m (9 ft) ceiling, and drop it in a hamper
- Design load 0.9 kg (jeans); first test 0.45 kg (1 lb)
- Success: one garment from floor to hamper with no human touch; I approve each move on the phone
- Safe state: PAUSE holds position, and nobody stands under the robot

Timeline and Scope

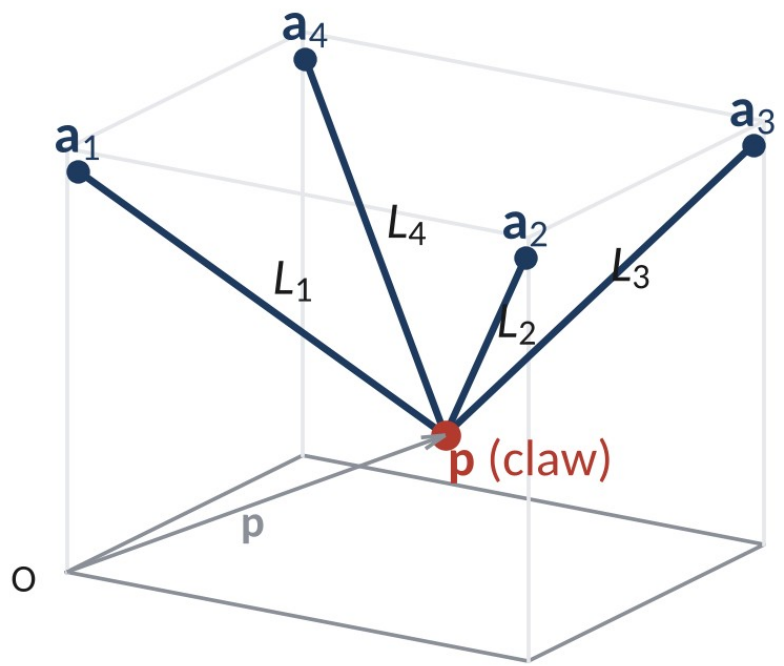
Phase	Dates
1. Concept	Jul 16 – Aug 5
2. Generation 1	Aug 6 – Sep 2
3. Redesign	Sep 3 – Oct 6
4. Final design	Oct 7 – Oct 21
5. Build and test	Oct 22 – Oct 27



1 Sep 30: first clamp's jaws cannot close on thin cloth 2 Oct 20: v6 clamp is 4.6 times its mass budget in CAD 3 Oct 22: drivers at 1.5 A; claw weighed

- Mine: requirements, design decisions, the SolidWorks station and claw, printing, machining, wiring and bring-up
- With Claude Code, under my direction: the control software, calculation scripts and CAD checks
- Purchased: steppers, drivers, controller boards, servo, camera, battery and encoder boards

Why a Cable Robot



	Floor robot	Gantry	Cable robot
Whole floor	Furniture blocks it	Yes	Yes
Floor left clear	No	Frame spans the room	Yes
Moving mass	Whole robot	Heavy beam	0.6 kg claw

- Chosen: four ceiling winches and one claw
- The cost: cables only pull, so cable tension sets the working space

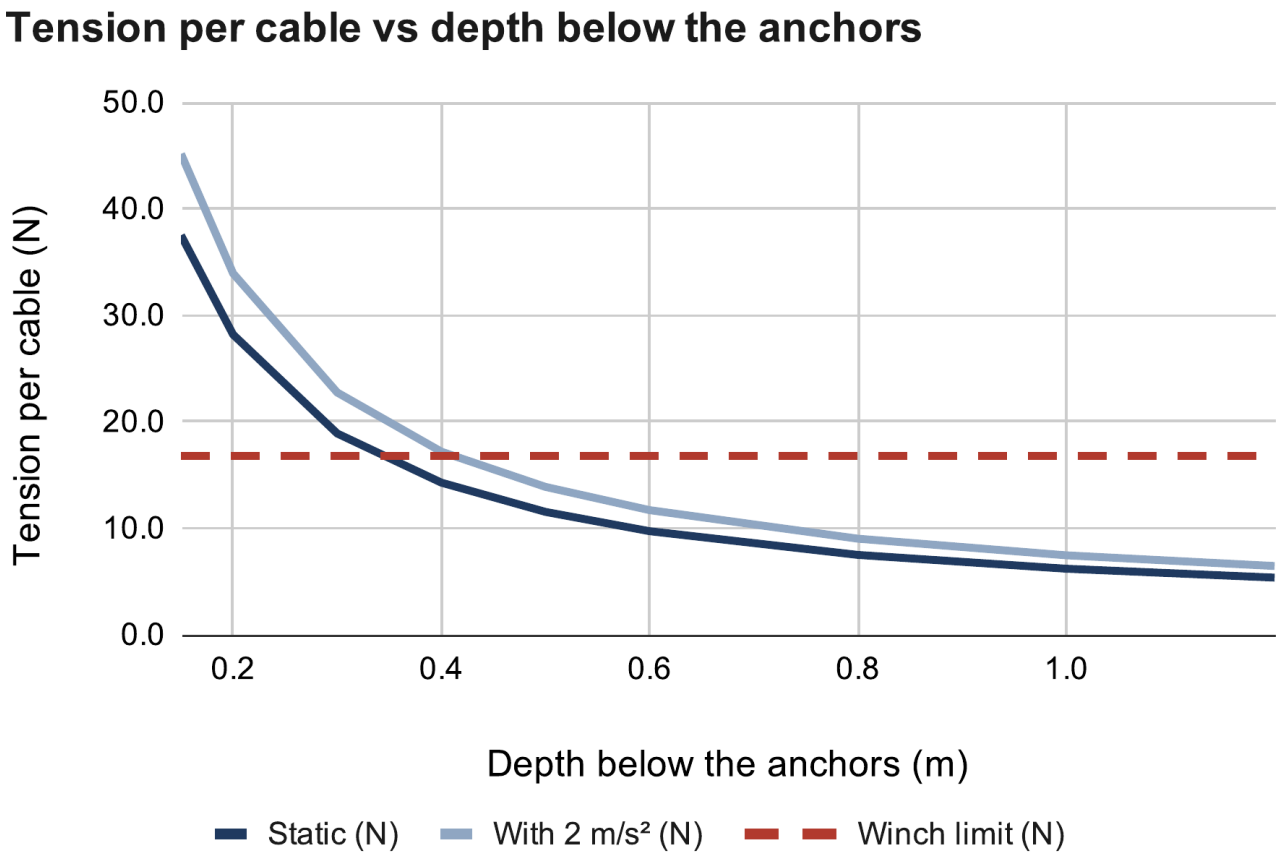
Cables from the ceiling anchors a_1 to a_4 to the claw at p

Design Trade-offs

Lever	Gain	Cost
Motor current up	Pull rises with current	Heat rises with current squared
Stronger claw	Stronger gears, firmer grip	More mass, less payload
Larger working area	More floor covered	Longer wires, more voltage drop
Closer to the ceiling	More working volume	Cable tension rises steeply
Smaller spool radius	More pull per amp	Tighter line bend, less room for the set screw
Larger safety margins	No skipped steps, no wall contact	Less usable force and floor

- The chain that binds: a heavier claw needs more pull, more pull needs more current, and heat rises with the square of the current

Cable Tension Sets the Ceiling Margin



Calculated (calc sheet: Cable tension)

- Weighed claw plus 1 lb, centred 0.4 m below the anchors: 14.3 N, 17.2 N at 2 m/s²
- That is at the 16.8 N winch limit; with jeans the claw must stay at least 0.6 m down
- The planner checks every cable at every path point against the measured pull limit

Tension per cable, 1 lb test at 0.4 m

$T = m\,(g + a) \,/\, (4 \sin \varphi)$

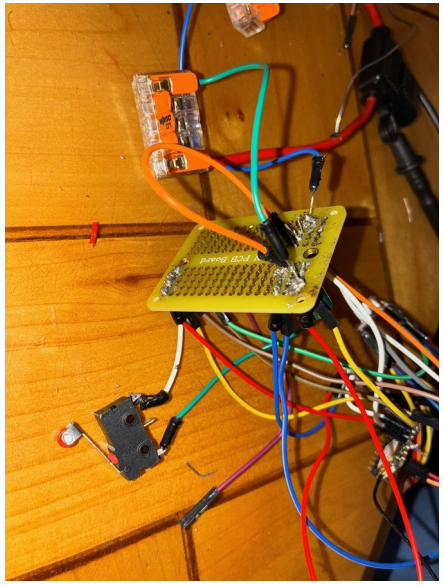
$= 1.054 \times 11.81 \,/\, (4 \times 0.181) =$ **17.2 N**

Winch limit: **16.8 N**

Generation 1: Bench Build and Findings



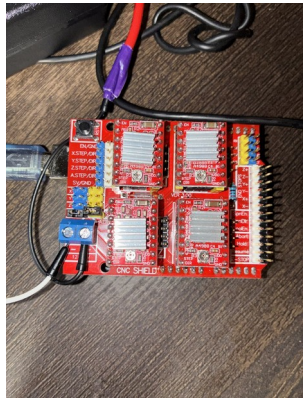


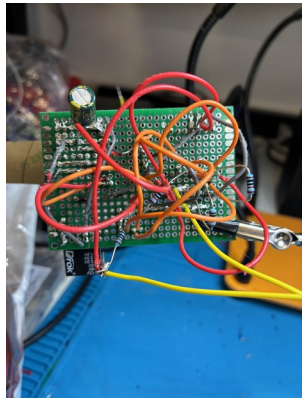


- A bench station lifted a test weight; the cable math held
- Bead homing would leave only 5–29 % of the room reachable, caught on paper before it was built
- One central board would need 10 wires to every corner once each winch had sensors
- Finger claw: designed in CAD (419 g of a 450 g budget), never built

Bench lift test, one-piece corner plate, homing switch

One Controller per Winch



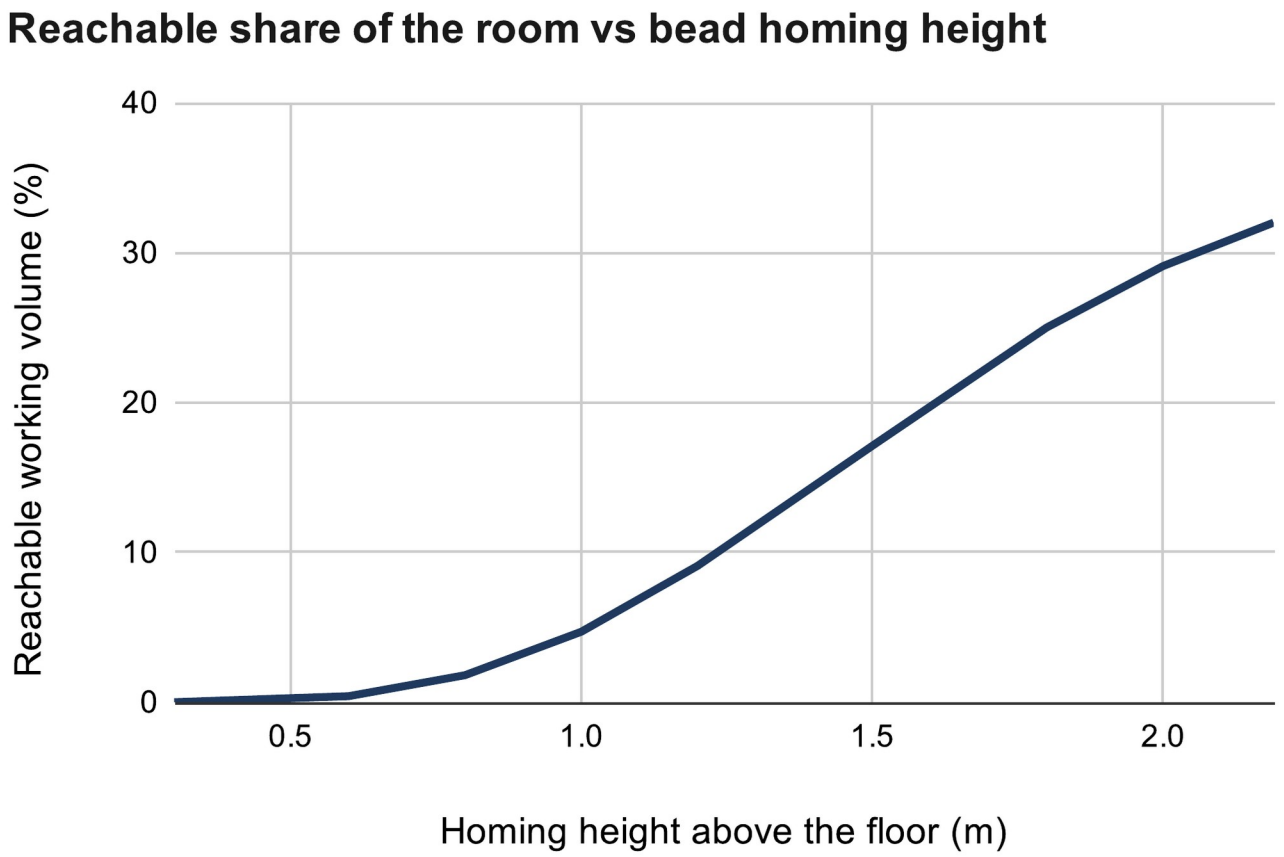


Before: one central driver board. After: a controller in each station.

	Generation 1	Final
Where is the claw?	Assumed from commanded cable lengths	Measured by the claw camera at every startup
Room geometry	Anchors from a tape measure	Anchors and cable lengths fitted, with error bars
Motor control	One central board	A controller per station, coordinated by the laptop
Finding laundry	Ceiling camera	Claw camera
Did the grab work?	Assumed	Camera checks pickup and delivery
Faults	Emergency stop only	PAUSE keeps torque; the emergency stop cuts power

- Generation 1: one central board; with sensors it would need 250 m (820 ft) of wire
- Now: a controller in each station, fed by one 12 V pair: 50 m (164 ft)
- Each station can be built and tested on its own
- Cost: timing moves into software; 50 ms of skew is 2.5 mm of cable error, about 7.5 N

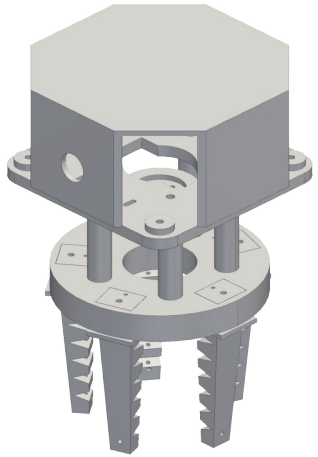
Camera Calibration Replaces Bead Homing



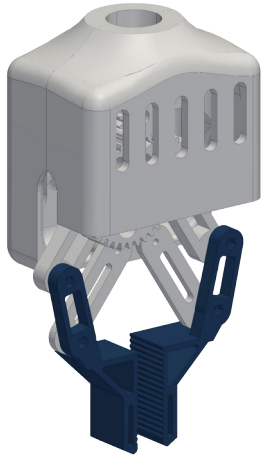
- Bead homing fixes each cable's shortest length, so most of the room is out of reach
- Instead the claw camera reads a printed floor board, and a fit finds the anchors and cable lengths
- Camera pose per frame, best case: 6.6 mm with one tag in view, 0.7 mm with four
- Simulated: cable lengths known to about 12–16 mm after the fit; the whole room opens only when the error bars allow it
- Beads stay on the cables as end stops: a bead reaching its switch stops that station

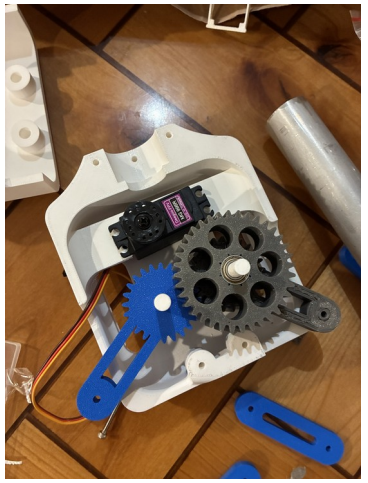
Calculated (calc sheet: Bead homing)

Claw: Four Versions to a Parallel Grip









Finger claw (CAD only), v6 clamp (CAD), final four-bar (CAD), printed gear train

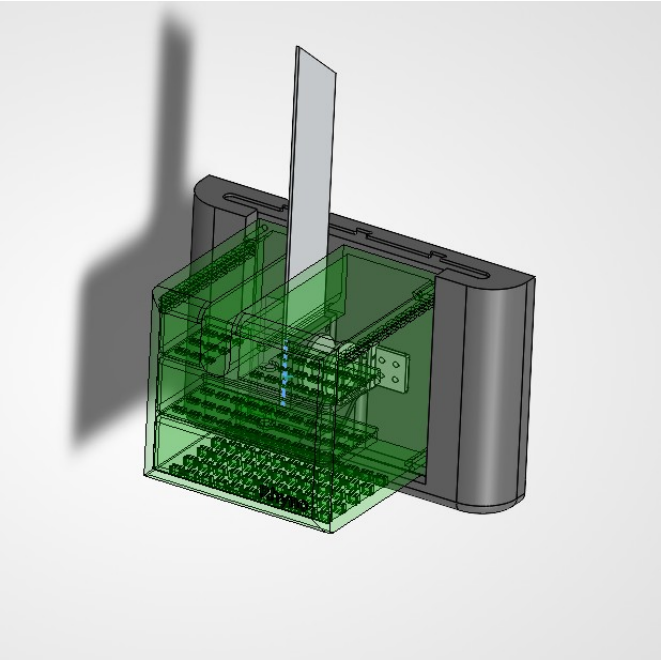
- Fingers (CAD only), then a clamp whose jaws could not close on thin cloth (Sep 30)
- v6 clamp: 4.6 times its mass budget in CAD (Oct 20)
- Final: geared four-bar, parallel jaws, 53 mm opening; gear safety factor 2.05 at servo stall
- Grip for jeans is short at narrow grabs
- Power: two-cell 7.4 V pack, 6 V servo rail, separate 5 V logic; peak 2.8–3.1 A at the pack's 3 A rating

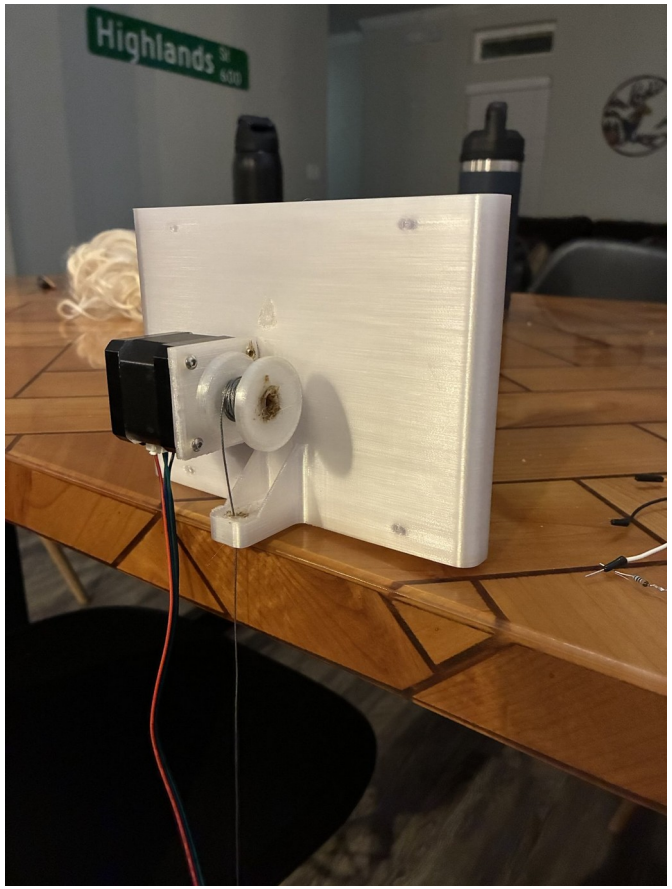
Grip force per jaw, jeans

$N \geq m\,(g + a) \,/\, (2\mu) = 0.9 \times 11.81 \,/\, 0.6 =$ **17.7 N**

Jaws: **13.3 N** narrowest to 25 N at a 70 mm grab

Final Wall Station





SolidWorks, shell transparent (left); the built station (right)

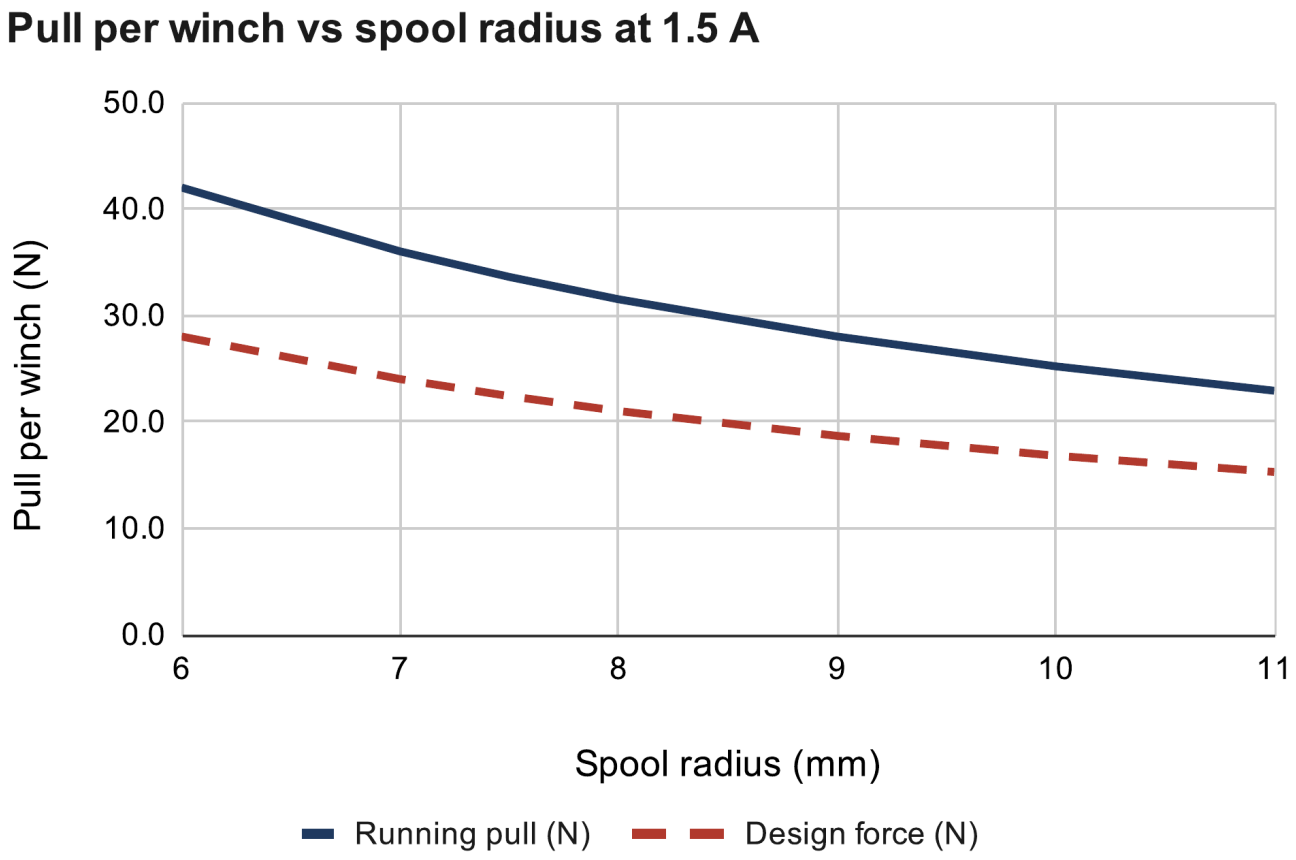
- Slide-in wall mount: the station lifts off while the anchor stays in the stud
- Load path: cable, spool, set screw, motor, PETG tab, slide-in, wall mount, screws, stud
- Design idea: the motor is the weakest link. It slips at 34–59 N, before any part breaks, and the spool encoder reports the overload
- Aluminum fin exits through a slot in the shell top

Motor slip force (the fuse)

$F = \tau_h / r = 0.36\ \text{N·m} / 0.0075\ \text{m} = \mathbf{48\ N}$

$F_{max} = 48\ \text{N} \times 1.11\ (\text{motor}) \times 1.11\ (\text{cable angle}) = \mathbf{59\ N}$

Winch Pull at 1.5 A



Calculated (calc sheet: Winch pull)

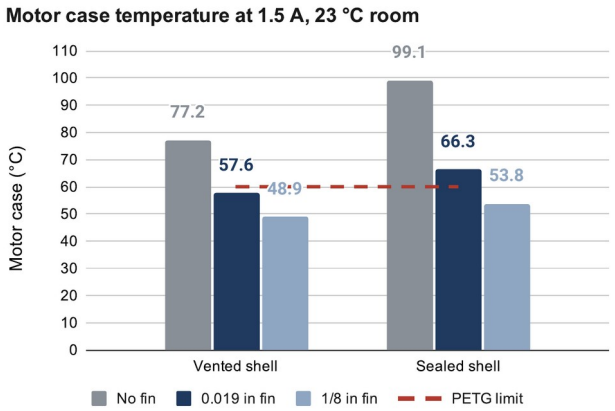
- NEMA 17 (National Electrical Manufacturers Association) stepper at its rated 1.5 A, up from 1.06 A: 41 % more pull
- Calculated pull: 34 N on an empty spool, 25 N when full
- Assumes running torque of 0.7 × holding torque; the pull test measures it

Winch design force

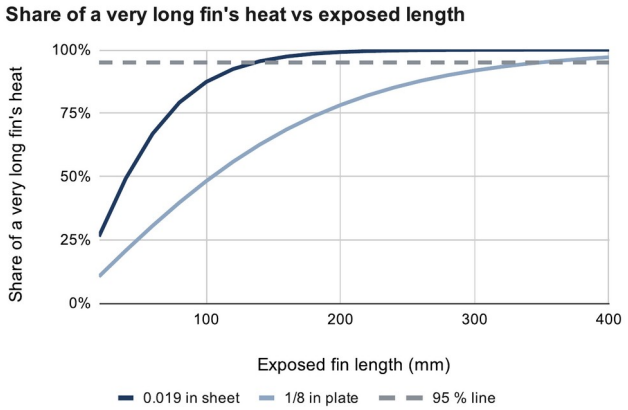
$T = \tau / r = 0.252\ \text{N·m} / 0.010\ \text{m} = \mathbf{25.2\ N}$

$F_{design} = 25.2\ \text{N} / 1.5 = \mathbf{16.8\ N}$

Motor Cooling



Calculated (calc sheet: Motor temperature; assumed 1.8 Ω coil, 23 °C room)



Calculated (calc sheet: Heat fin; Incropera, ch. 3)

- 1.5 A doubles motor heat to 4.05 W; the PETG mount must stay below 60 °C
- Vented shell and fin: 56–59 °C with the 0.019 in prototype, 45–49 °C with the 1/8 in design
- The thin sheet stops gaining by 136 mm; the 1/8 in plate keeps gaining to 349 mm

If the fin alone carried all 4.05 W (painted 6061, 44 mm wide)

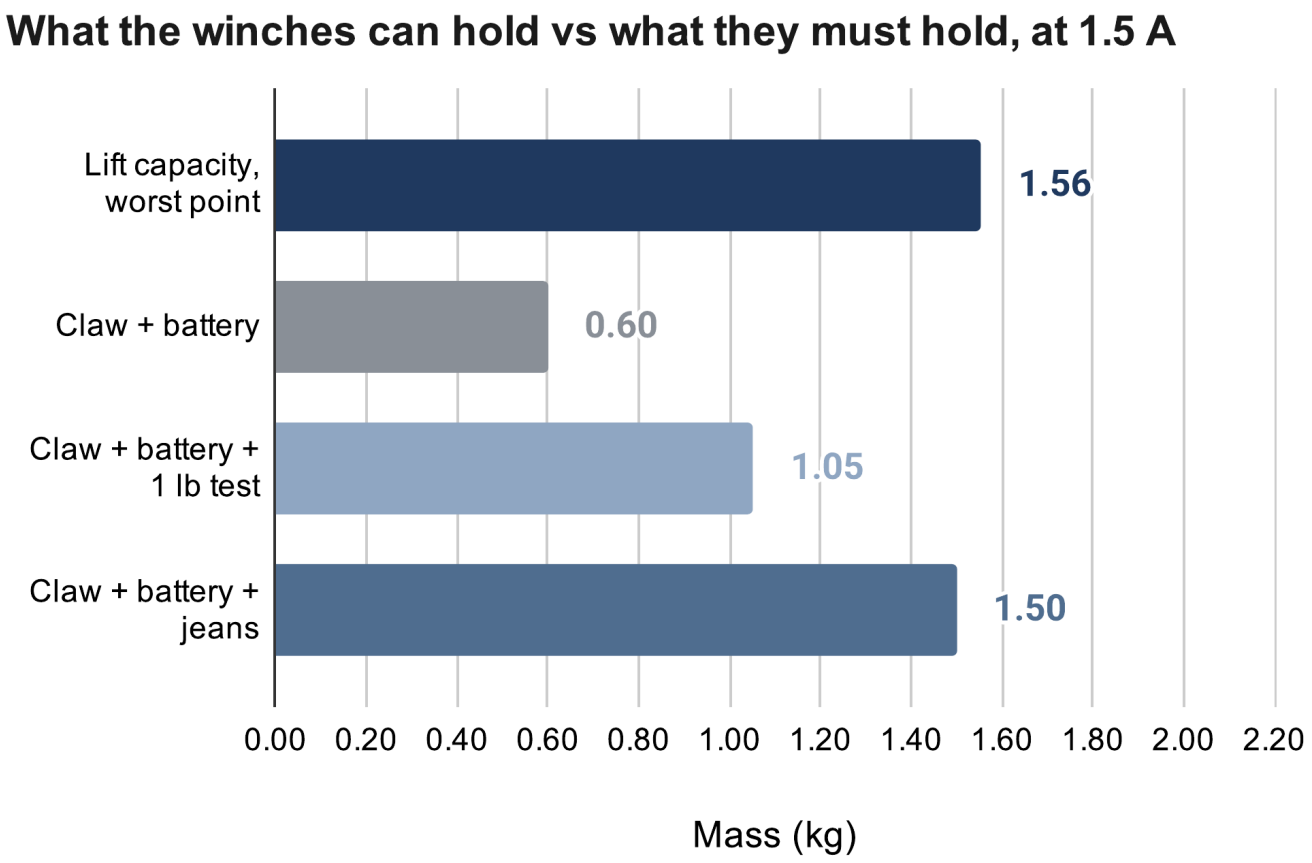
Thickness	Mass	Shortest for ≤ 60 °C
0.019 in (prototype)	14 g	None (2.0 W max)
1/16 in	47 g	None (3.6 W max)
0.090 in	68 g	210 mm or none
1/8 in (design)	94 g	145–185 mm
3/16 in	141 g	125–140 mm
1/4 in	189 g	115–125 mm

Motor heat, $P = I^2 R$

At 1.06 A: $(1.06\ \text{A})^2 \times 1.8\ \Omega = 2.0\ \text{W}$

At 1.5 A: $(1.5\ \text{A})^2 \times 1.8\ \Omega = \mathbf{4.05\ W}$

Weight Budget



Calculated (calc sheet: Lift budget)

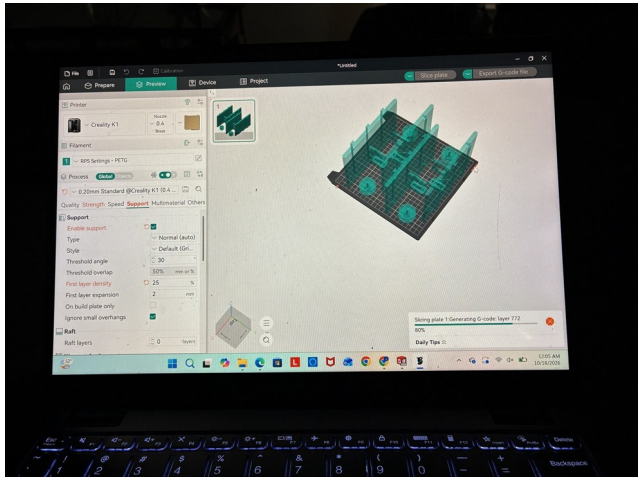
- Claw with battery weighed 0.60 kg (0.86 kg estimated)
- Calculated capacity at task height 1.56 kg; jeans need 1.50 kg, a 4 % margin
- That margin is inside the uncertainty: at 0.5 × holding torque, capacity is 1.1 kg
- The pull test decides: keep 1.5 A, or add a 2:1 gear stage at each winch

Lift capacity

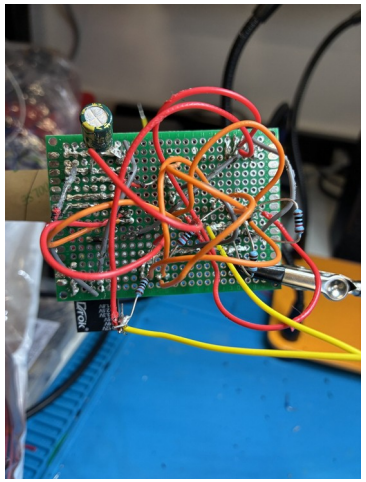
$$m_{max} = F_{design} / (k\,g) = 16.8 / (1.101 \times 9.81)$$

$$= \mathbf{1.56\ kg} \text{ against } 0.60 + 0.90 = \mathbf{1.50\ kg}$$

Build: Printing, Machining, Bring-Up





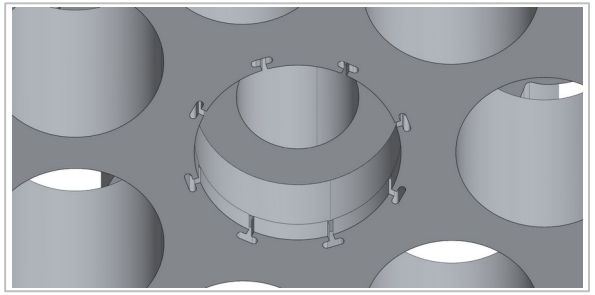




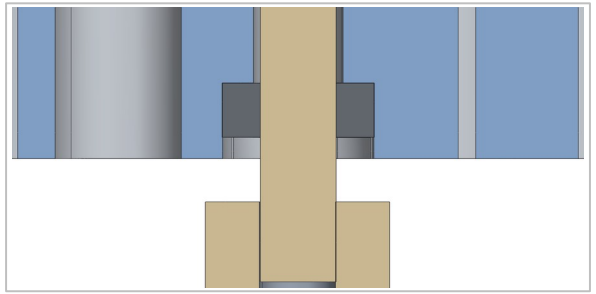
Slicing PETG for the K1, finished fin, soldered station board, mill setup

- One spool bore printed too tight (Ø5.2 on a 5 mm shaft); v3 opened it to Ø5.45
- Fin drilled in one mill setup from a 1:1 drawing; the 7/8 in pilot fits the 22 mm motor boss
- First motor would not turn: logic rail at 1.14 V, traced to a bad 3.3 V connection
- Now: a meter on every rail before code, and fit coupons before new fits

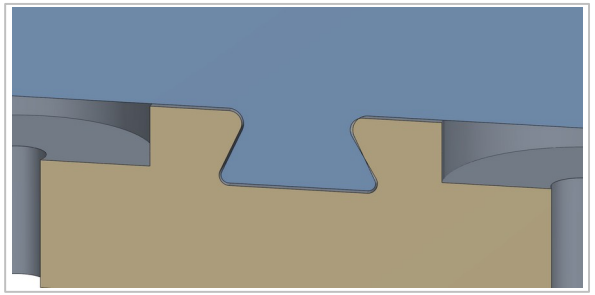
Manufacturing: Fits and Datums



Gear: Ø14.000 entry lip, 8 T-slots



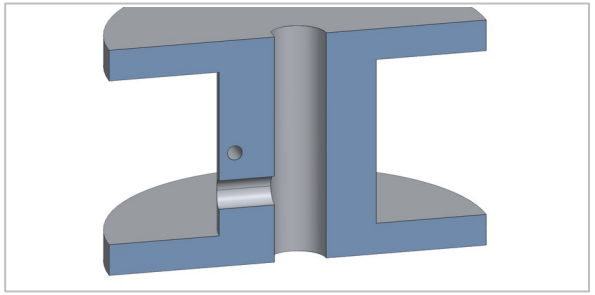
Cut open: 687 bearing on the printed pin



Slide-in rib in the wall-mount dovetail



End-stop switch pressed into the slide-in



v3 spool: D-bore and set-screw hole



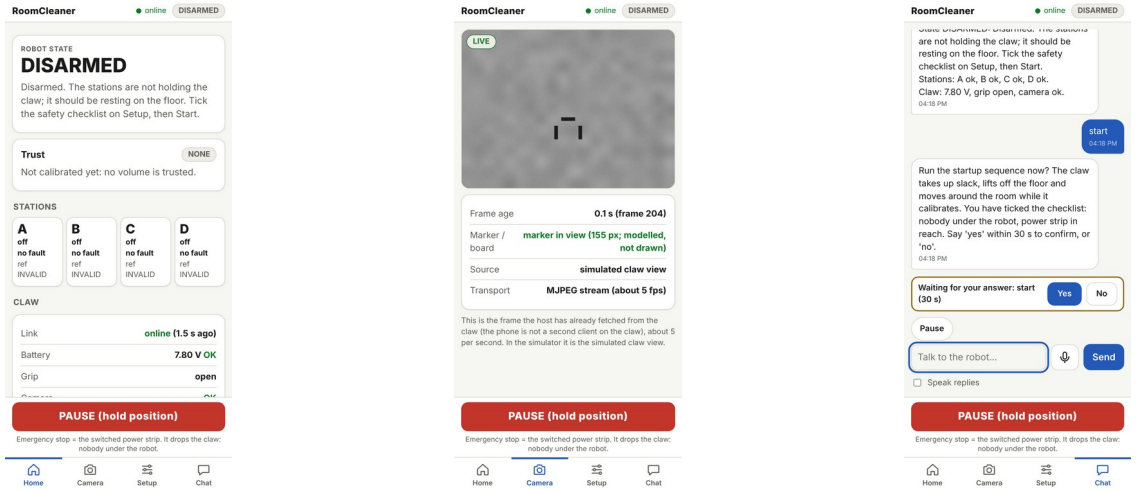
Fin: holes drilled from the first hole

My SolidWorks parts; blue and tan are cut faces, dark grey the bought bearing

Feature	As drawn	Fit
Gear bearings (687)	Ø14.075 pocket behind a Ø14.000 lip, 1.95 long, 8 T-slots	Snaps past the lip, +0.075 behind it
Gear pins	Printed Ø7.000 pin into a Ø7.05 socket	Line-to-line in the bearing; +0.05
Slide-in mount	0.175 per side; 30° dovetail flanks	Slip fit, 0.14–0.21 on the flanks
Spool on Ø5 D-shaft	Ø5.30 bore (10/21), Ø5.45 in v3; 0.5 mm flat	+0.30 to +0.45: Ø5.2 printed too tight
End-stop switch	Pocket 6.65 × 20.3 × 9 on a 1 mm ledge, lever window below	Press fit, no screws
Heat-set inserts	M3: Ø4.0 × 5 gear housing; M2.5: Ø3.45 × 4 encoder	Melt-in; hole deeper than the insert
Screw clearance	M3 Ø3.5 printed, Ø3.4 drilled; wall Ø3.97 (5/32 in)	+0.5 printed, +0.4 drilled
Motor pilot, Ø22 boss	Ø22.4 printed tab, Ø22.5 drilled fin	+0.4 / +0.5
Fin hole zero	Edges locate the bottom-left hole; drilled, it is the zero for the other 4	Pattern ±0.1, free of the ±0.5 sheared edges

- Printed holes come out small, so every printed fit carries an allowance; the drilled fin needs none
- PETG on a Creality K1, 0.4 mm nozzle, 0.2 mm layers; loads along the layers, mating faces off supports

First System Test



Phone app: home, live claw camera, chat; every move waits for approval

Check	Pass limit	Result
Self-check	All stations answer; slack taken up	Planned
Calibration	Held-out poses: cable residual ≤ 20 mm	Planned
Short move, 1 lb	300 mm at 50 mm/s; cable mismatch ≤ 2 mm	Planned
Pickup	One garment into the hamper	Planned

- Three stations hold the claw under gravity over the triangle they span, about half the floor
- Sequence: self-check, calibration, short move, then pickup
- 1 lb first, then one garment; pass limits set before the run

If something goes wrong

Event	What happens
PAUSE or a detected fault	All stations hold with torque on; nothing restarts without approval
Wi-Fi link lost	Stations stop and hold on their own clocks
Power loss or emergency stop	Torque is lost and the claw can fall, so nobody stands underneath; brake planned
Reset or brown-out	Driver off at boot: same as a power loss
Driver over-temperature	The A4988 shuts off; the hold test watches it

Results Against Targets

Target	Goal	Current	Evidence	Status
Lift capacity	1.50 kg with jeans	1.56 kg; 1.1 kg at 0.5 × holding torque	Calculated	Marginal; pull test
Grip for jeans	17.7 N per jaw	13.3–25 N	Calculated	**Short at narrow grabs**
Claw mass	0.45 kg	0.60 kg with battery	Weighed	**Not met**
Motor temperature	Below 60 °C	56–59 °C prototype fin	Calculated	Marginal; hold test
Weakest link	Motor slips first	Set screw may slip first; tab at 1.2 ×	Calculated	**Two fixes**
Working area	Floor, 0.2 m wall margin	All but 0.2 m at the walls	Calculated	Met in design
Wiring	One cable per station	One 12 V pair; 0.51 V drop	Built	Met
Accuracy	≤ 2 mm cable mismatch	40 ms skew budget gives 2 mm	Not measured	To be tested
Speed	50 mm/s	Not yet run	None	To be tested
Cost	Hobby budget	Not yet totaled	Receipts	Open

Good and Poor Decisions

Good decision	Result
One controller per winch	Five times less wire; each station tested alone
Camera calibration instead of bead homing	Whole room reachable; caught on paper before building
Teeth checked at servo stall	A clamp stalls the servo on every grip: safety factor 2.05

Poor decision	Result
Mass not tracked through four claw versions	v6 reached 4.6 times its budget in CAD; the claw is 0.15 kg over
Prototype fin cut from scrap sheet	Only 1–4 K of thermal margin
Build and test week ends on presentation day	System results arrive at the last minute
Parts built far stronger than any load needed	Station mount 17 to 225 times the motor's pull; 1.2 L of PETG per station

- Thorough where a mistake is expensive: fin holes drilled from the first hole as zero, the stall check on the teeth
- Scrappy where it is cheap to undo: a jumpered ESP32 after a controller failed, a floor board printed on paper

Takeaways and Next Steps

Takeaways

- Pull-test one winch and weigh a mock claw in week 1; set the mass budget from that
- Kill ideas on paper first: bead homing and the central board failed calculations I could have run in week 1
- Prove one full station, calibration included, before copying it
- Size every part to a load number; over-building cost weight, plastic and print time

Next steps

- Pull test, then enter the measured limit in the planner
- Hold test at 1.5 A, three-station test, grip pads
- Spool set screw into metal; 6 mm motor tab with fillets on both roots
- Lighter redesign sized to the loads: about 60 % less station plastic, claw to its 0.45 kg budget
- Continuous testing: long runs logging tracking faults, temperatures and creep
- A brake or non-backdrivable stage, so a power loss cannot drop the claw